

DSC 411/511
Advance Data Mining Algorithms and Applications

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Location: MSB 104



Lecture 01
Introduction to Data Mining

Today's class:

- *What is data mining?*
- *Data mining vs Statistics vs Machine Learning*
- *The four big tasks*
- *CRISP-DM: the process we follow all semester*

Your First Day on the Job



The Scene

- You are hired as a data scientist at a grocery chain.
- They hand you 20 months of receipts. 2.4 million of them.
- No labels. No instructions. Nobody who remembers how it was collected.



- You have two weeks
- Find something useful

So where do you even start?

Here is What You Actually See

Raw Data

receipt_id	date	store	item	. . .	qty	price	member_id
1000001	2025-03-04	S12	Whole Milk 2L	. . .	1	3.49	M-88213
1000001	2025-03-04	S12	Diapers Size 3	. . .	1	24.99	M-88213
1000001	2025-03-04	S12	Beer 6pk	. . .	1	9.99	M-88213
1000002	2025-03-04	S07	Bananas	. . .	3	0.59	NULL
1000003	2025-03-05	S12	Bread Whole Wheat	. . .	1	2.79	M-41902
1000003	2025-03-05	S12	Eggs Large 12ct	. . .	2	4.15	M-41902
...	...	...	...	. . .	...	...	...

2,399,993 more rows

The Human Limit

- Reading 2.4 million rows at one second each takes 28 days without sleep.
- Excel gives up at 1,048,576 rows. Your file is more than double that.
- Sorting helps you find one thing. It does not find patterns.
- The useful patterns sit across columns, not inside any single column.

Nothing in here tells you what to look for.

What is Data Mining?

Textbook Definition 1:

Data mining is the process of automatically discovering useful information in large data repositories.

Introduction to Data Mining, 2nd Edition

Textbook Definition 2:

Data mining is the process of discovering interesting patterns and knowledge from large amounts of data.

Data Mining: Concepts and Techniques, 4th Edition

Working Definition

- **Finding patterns in large data that are all three of these:**
  Novel: you did not already know them.
  Valid: they still hold on data you have not seen yet.
  Useful: somebody can actually act on them.

Can you give an example of pattern for the receipt data?

What Does a Pattern Look Like?

Three Real Findings From Those Receipts

1. **Seasonal trend**

   Ice cream and cold drinks jump 60 percent in June, July and August.
   Use it: stock up before June, cut the orders back in September.



2. **Association rule**

   85 percent of baskets holding bread and eggs also hold milk.
   Use it: move milk next to the bread aisle, or bundle all three.



3. **Customer behavior**

   Weekend-only members spend twice as much per visit as weekday shoppers.
   Use it: run weekend promotions and put more staff on those days.



Data Mining vs Statistics vs Machine Learning

Same Grocery Data, Three Different Questions

- **Statistics asks:**
  "Do members really spend more than non-members? Prove it."
  (You bring the hypothesis, the data confirms or rejects it.)

- **Machine Learning asks:**
  "Given a shopper's history, predict what they buy next."
  (*You know the goal, you want accuracy.)*

- **Data Mining asks:**
  "Here is everything we have. Find anything worth knowing."
  (Find anything worth knowing.)

Same tools. What differs is how much you know before you start.

Where All This Data Comes From

The Numbers Are Hard to Picture

- **Walmart: more than 1 million customer transactions every hour.**
  Those are receipts. Exactly like the ones on your screen a minute ago.

- **One self-driving car: 1 to 5 TB of sensor data per hour.**
  A single day of testing fills more storage than your whole laptop.

- **One 1080p CCTV camera: roughly 40 GB per day.**
  A campus with 100 cameras produces about 4 TB. Every single day.

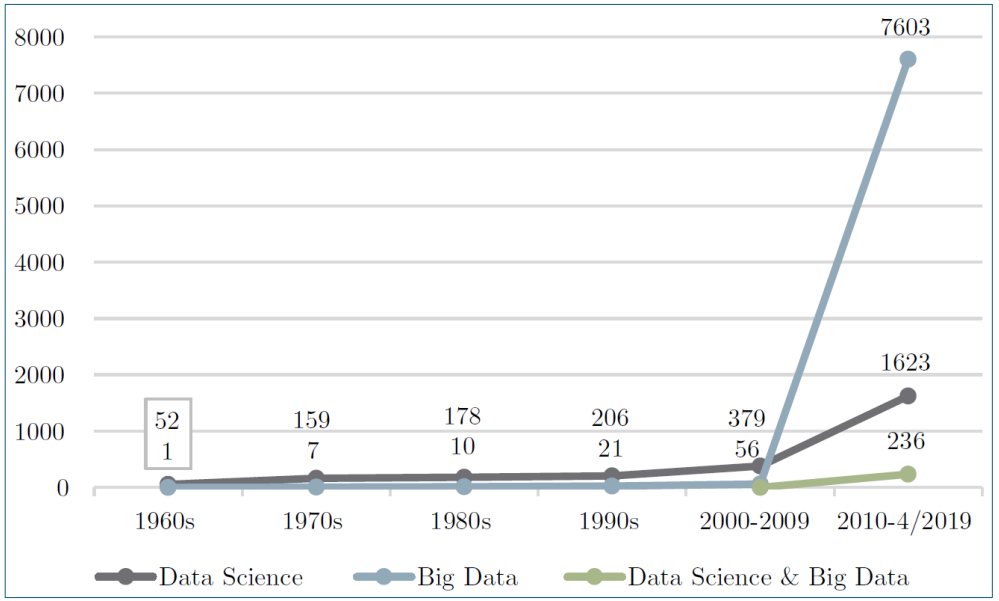
- **WhatsApp: over 100 billion messages a day.**
  That is more than 1 million messages every second.

Why Is This Happening Now?

Three Things Changed

- Storage got cheap. Keeping data costs less than deciding what to delete.
- Computing got cheap. A laptop today beats a 1990s supercomputer.
- Everything became a log file. Every tap, swipe and scan is recorded.

Settings → Privacy & Security → Analytics & Improvements → Analytics Data



Evolutionary trend in the number of publications covering data science and big data

The Four Big Tasks

Almost Every Problem is One of These

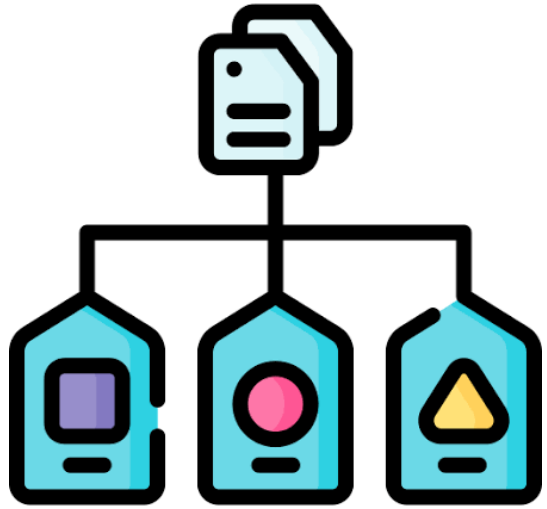
- Classification: predict a label you already know exists.
- Clustering: group things when you have no labels at all.
- Association: find what keeps showing up together.
- Anomaly Detection: find the few that do not fit.

The hard part is not the code.
It is working out which of the four your problem actually is.

The Four Big Tasks

Task 1: Classification (Predict a Label)

- You start with examples that are already labeled.
- The model learns the pattern, then labels new cases for you.



Grocery example:

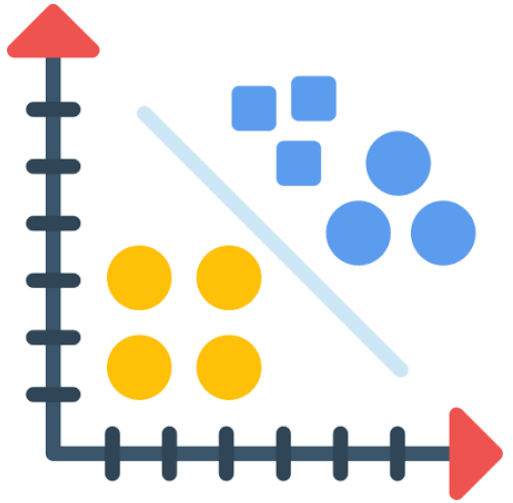
"Will this member stop shopping with us in the next 3 months?"
Labels: Stayed or Left. Learned from last year's members.

- Also: spam filters, loan approval, medical diagnosis.

The Four Big Tasks

Task 2: Clustering (Group Without Labels)

- Nobody tells you what the groups are. The algorithm finds them.
- Then you look at each group and work out what it means.



Grocery example:

The algorithm splits shoppers into 5 groups on its own.
You look and name them: bulk families, snackers, weekend cooks.

- Careful: it will always hand you groups. Even in pure noise.

The Four Big Tasks

Task 3: Association (What Goes With What)

- Looks for items that keep landing in the same basket.
- The output is a rule: if someone buys A, they often also buy B.

Grocery example:

"Shoppers who buy diapers on Friday evening also buy beer."
Famous story. Probably a myth. We will test it ourselves later.

- Used for shelf layout, bundle deals, and recommendations.



The Four Big Tasks

Task 4: Anomaly Detection (Find the Odd One Out)

- Most of your data is normal. You want the tiny slice that is not.
- The catch: you rarely have many examples of what “weird” looks like.



Grocery example:

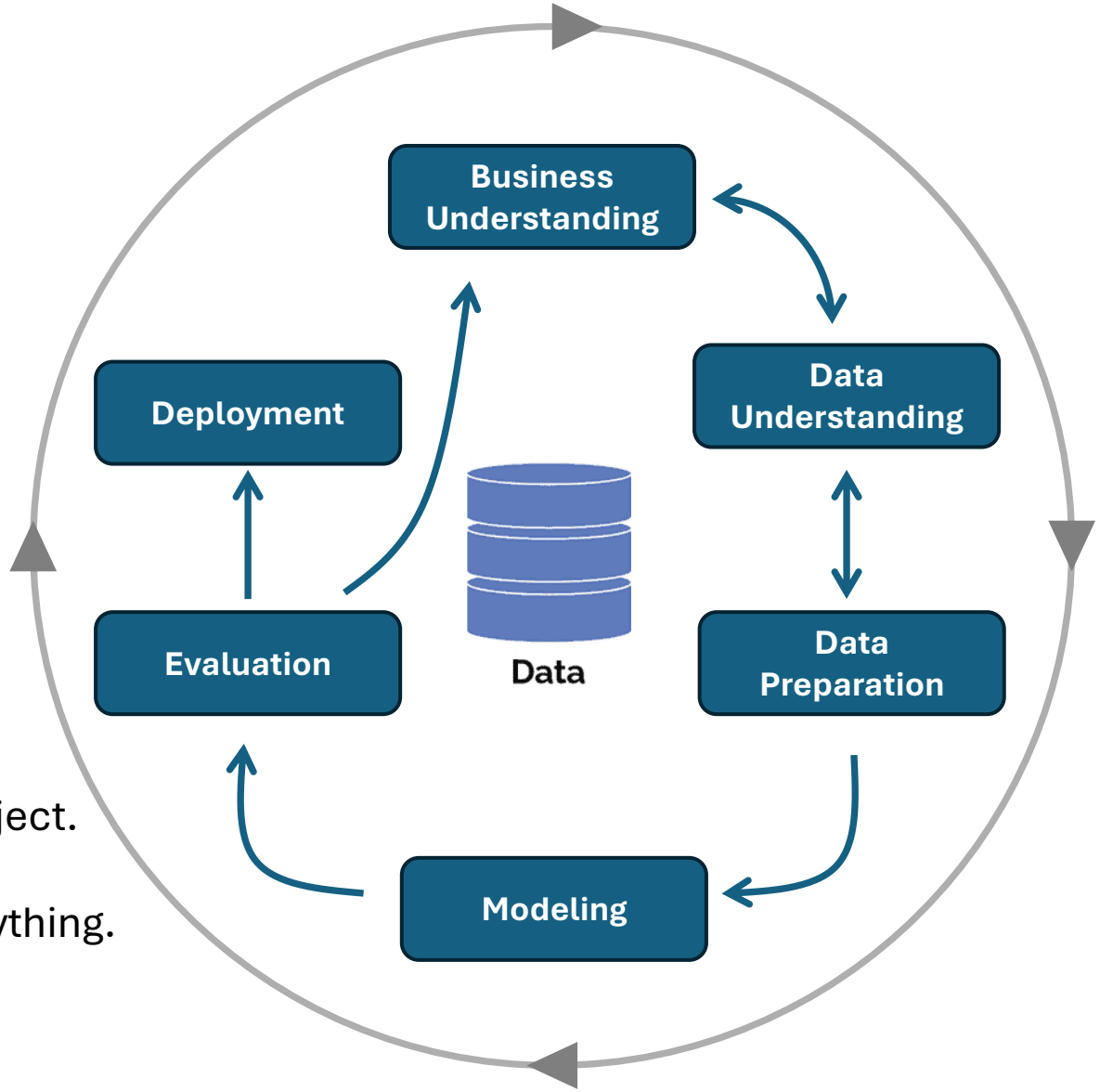
One store logs 400 returns in a single hour at 2 in the morning. Could be fraud. Could be a broken scanner. Both worth a look.

- Used for fraud, intrusion detection, machine failure, quality control.

CRISP-DM: How We Will Work

CRoss
Industry
Standard
Process for
Data Mining

- CRISP-DM is a process model for a data mining project.
- The algorithms live in step 4.
- The other five steps are what make step 4 worth anything.



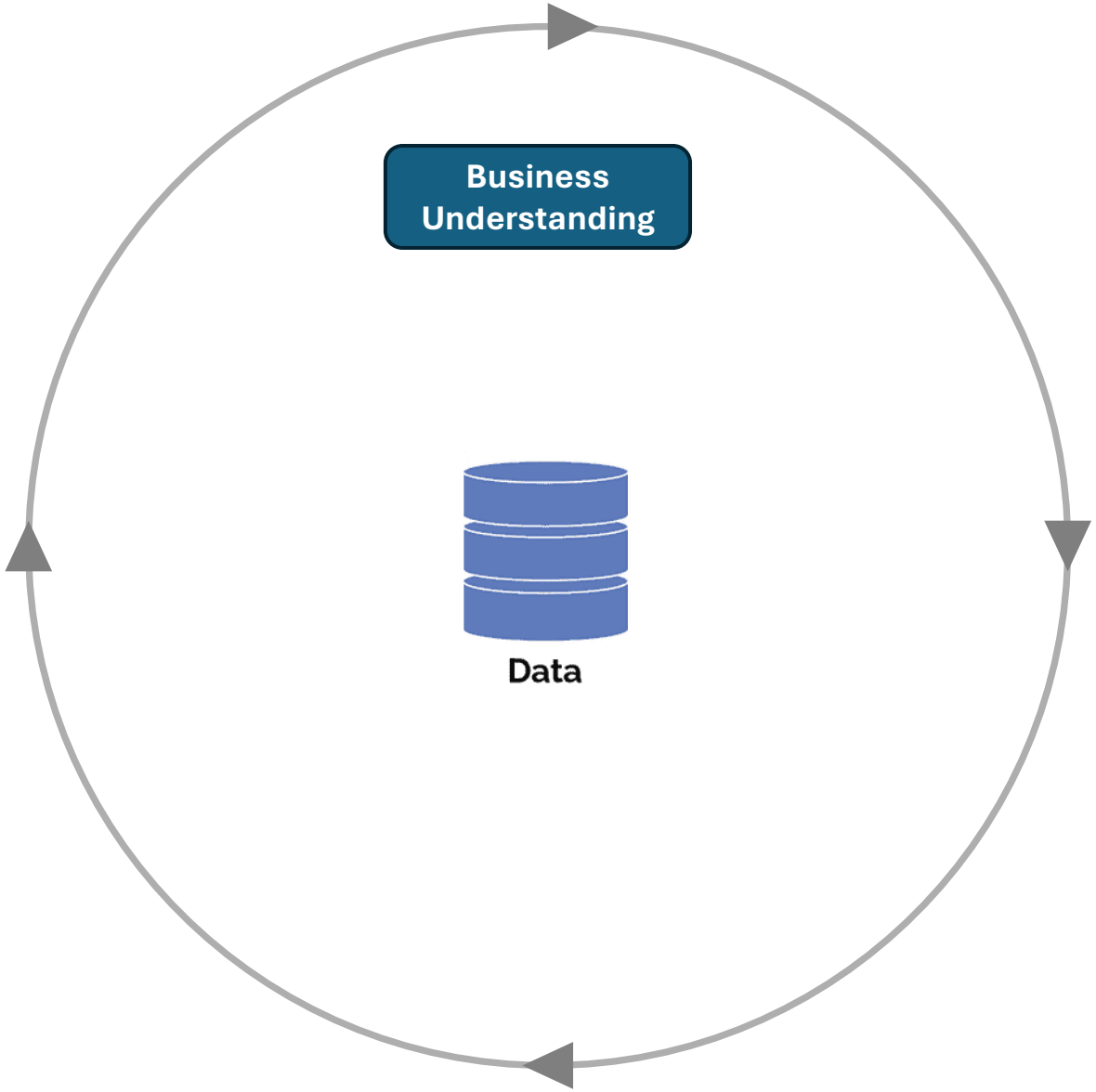
CRISP-DM: How We Will Work

1. Business Understanding

- Determine business objectives
- Assess the situation
- Determine data mining goals
- Produce project plan

Grocery example:

- Ask: “Why are we losing members?”
- Not “find something useful.”



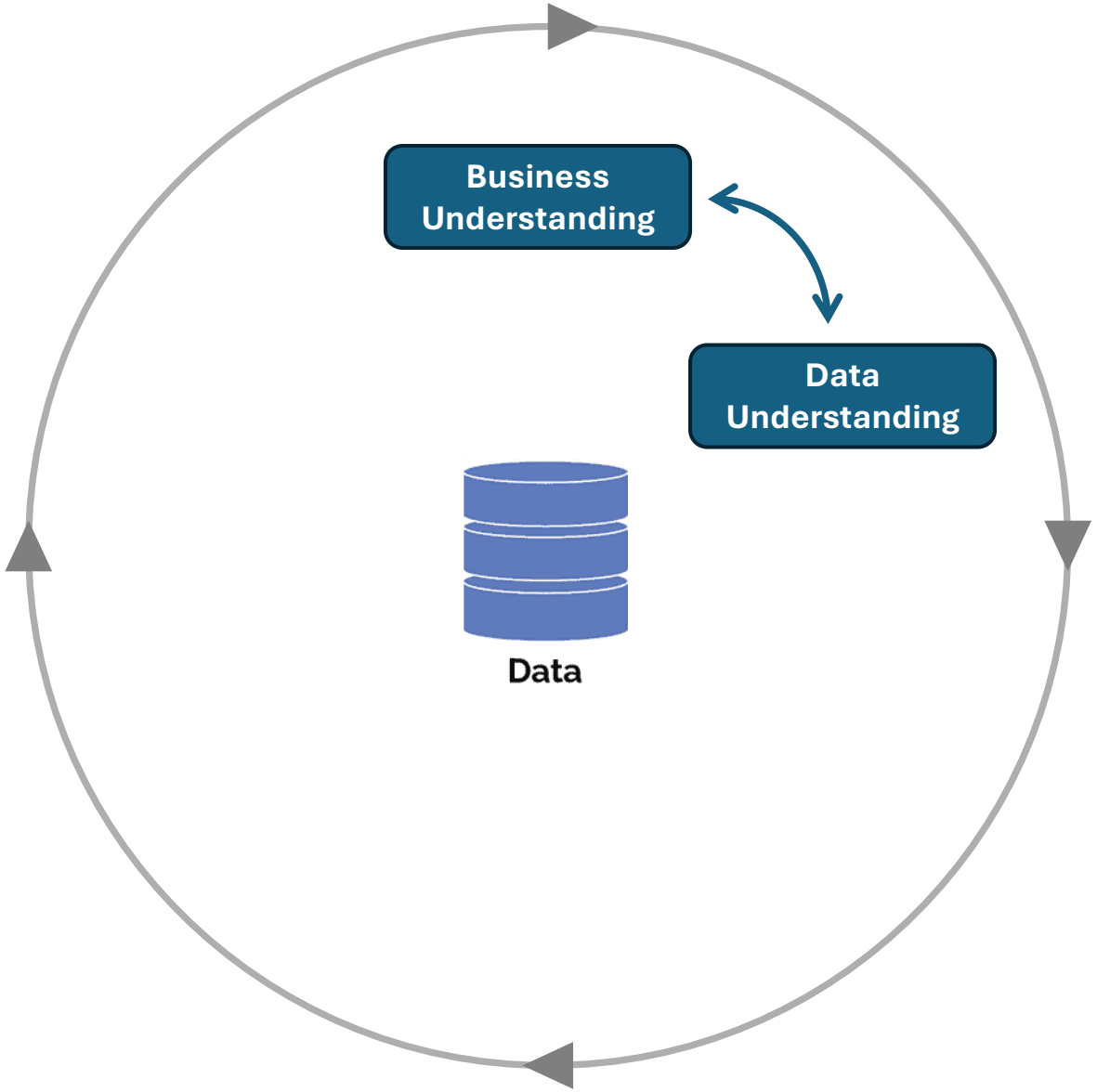
CRISP-DM: How We Will Work

2. Data Understanding

- Collect the initial data
- Describe the data
- Explore the data
- Verify data quality

Grocery example:

- 2.4M rows and 20 columns
- 12 percent of member_id are NULL



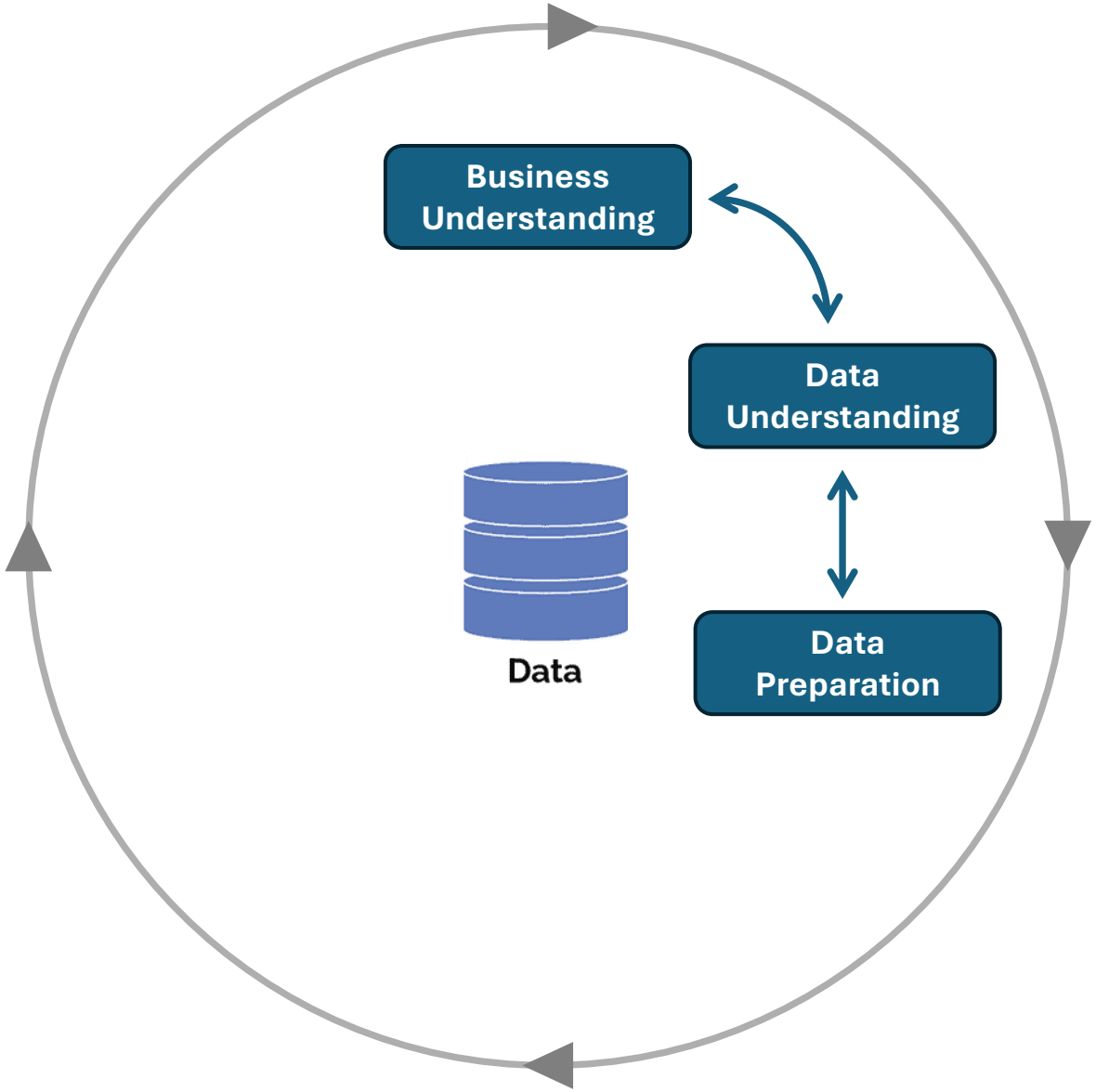
CRISP-DM: How We Will Work

3. Data Preparation

- Select data
- Clean data
- Construct data
- Integrate data
- Format data

Grocery example:

- Drop the NULLs
- Build spend per member
- Months since last visit



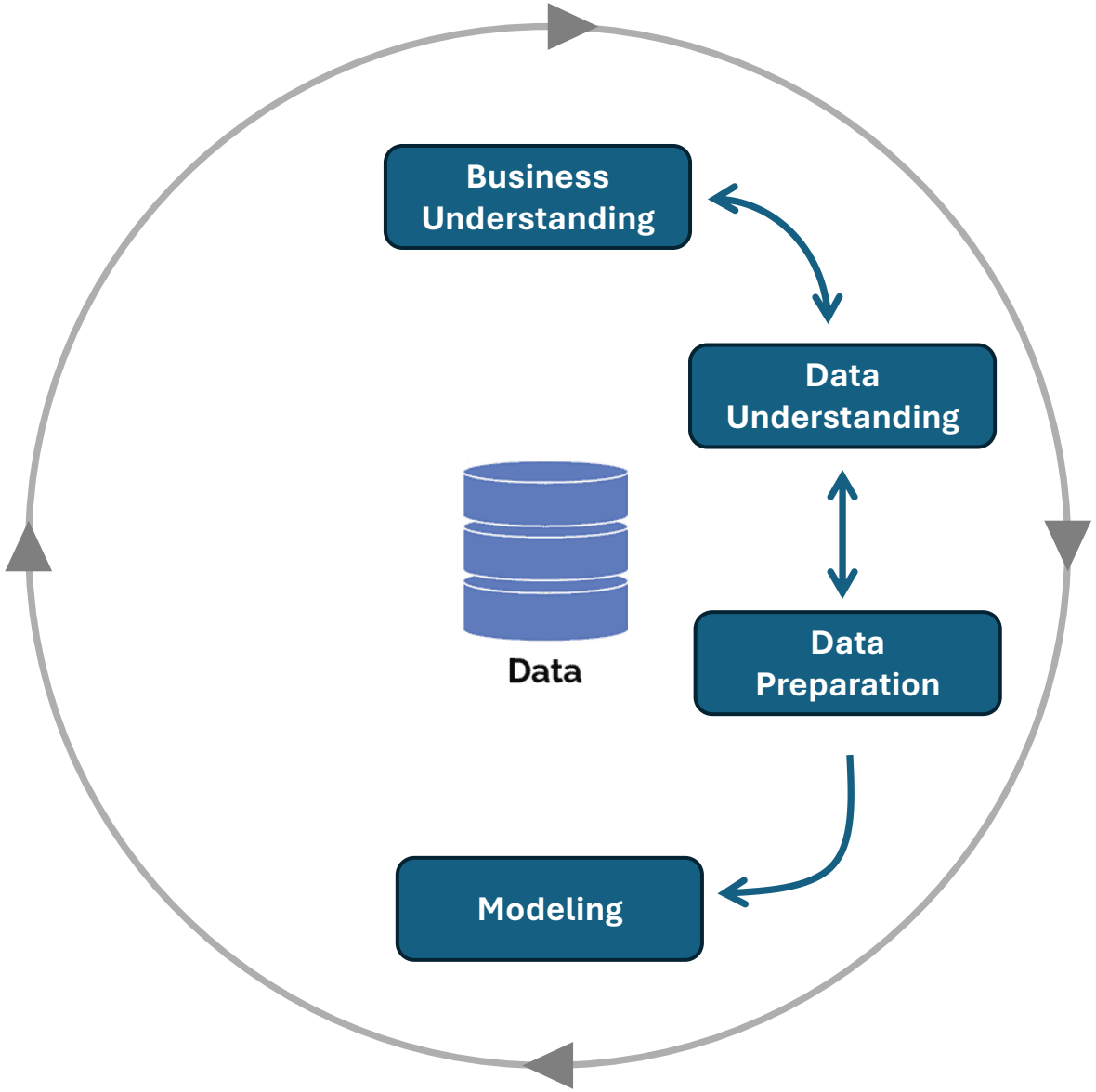
CRISP-DM: How We Will Work

4. Modeling

- Select the modeling techniques
- Generate test design
- Build model
- Assess model

Grocery example:

- Cluster the members
- Then classify who is about to leave.



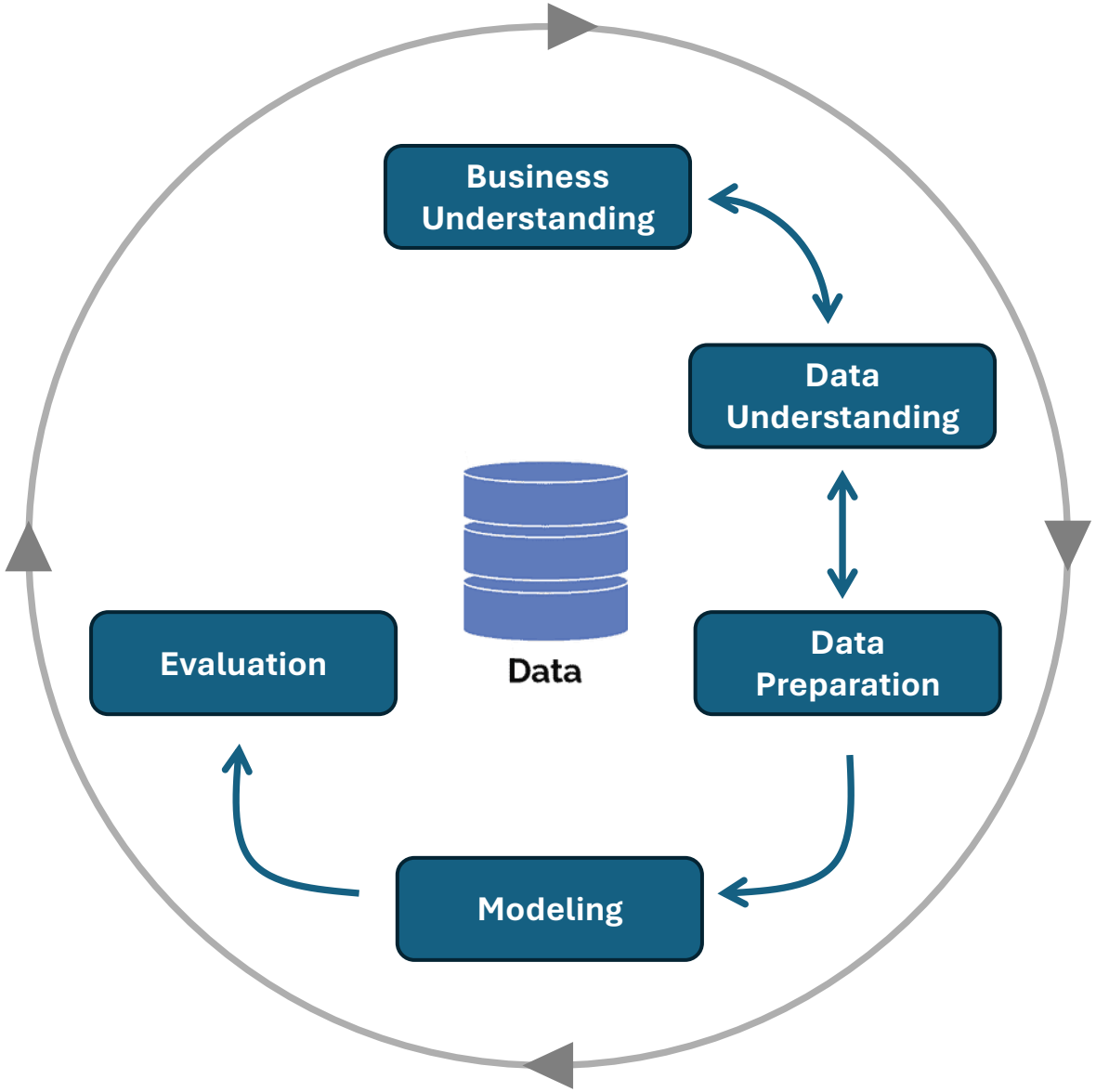
CRISP-DM: How We Will Work

5. Evaluation

- Evaluate the results
- Review process
- Determine next steps

Grocery example:

- does it catch real leavers, or just guess “nobody leaves”?



CRISP-DM: How We Will Work

6. Deployment

- Plan deployment
- Plan monitoring and maintenance
- Produce final report/results
- Review project

Grocery example:

- a weekly at-risk list that marketing can actually use

